Fenrix Strategy Validation Lab

Backtesting shows where a strategy worked. Fenrix searches for where it breaks.

- Product flow: Describe → Review → Lock → Backtest → Stress → Minimize → Export
- Strategy hash (immutable): c9d19b2e8d20667a…
- Source code SHA: 8b2140bdb9ff5333b6e457df79c20e43ac9f9738 · data mode of record: yfinance · universe: 30 assets

The problem

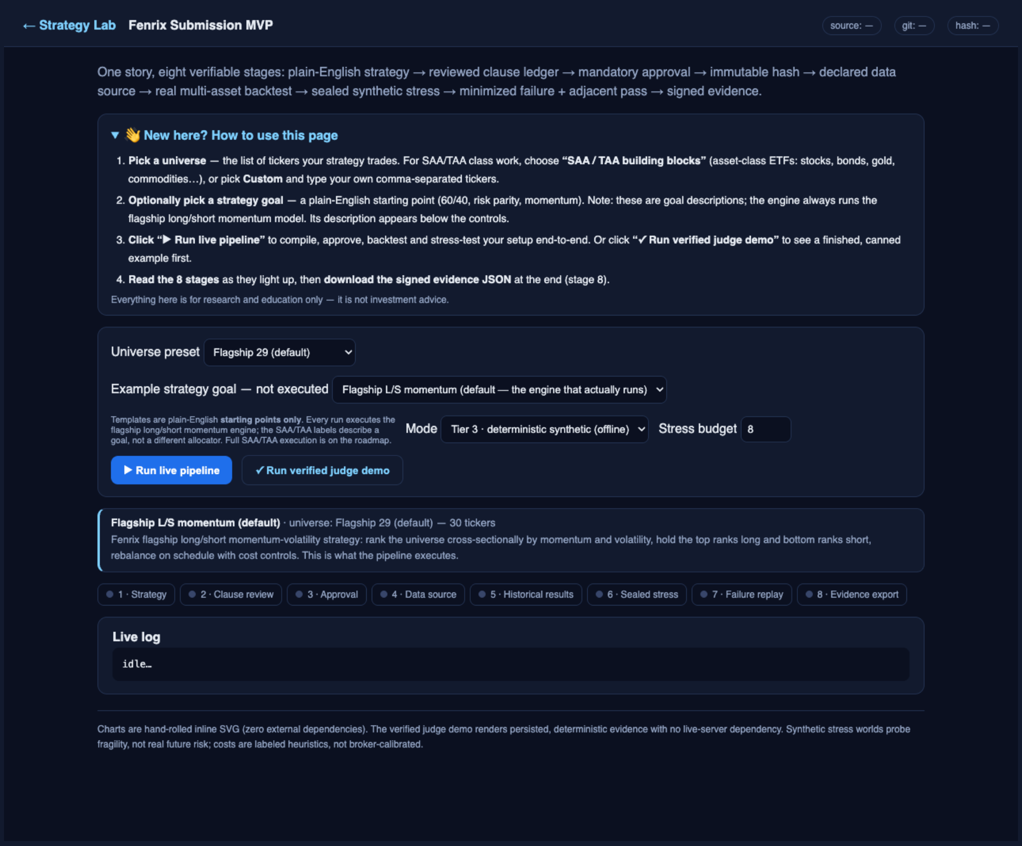
A green backtest is a claim, not a validation

- Backtests only tell you where a strategy already worked — they are silent about fragility.
- Students and retail quants ship strategies validated on one historical path.
- Overfitting, cost blindness and regime fragility hide until real capital finds them.
- There is no cheap, reproducible way to ask: under which conditions does this break?

Product workflow

Describe → Review → Lock → Backtest → Stress → Minimize → Export

- Describe: plain-English strategy → structured clause ledger (every clause reviewed).
- Review + Lock: mandatory approve step → immutable version + canonical SHA-256 hash.
- Backtest: the SAME locked hash runs the historical backtest on the run-of-record data (yfinance).
- Stress: the SAME hash enters a sealed synthetic failure search across mechanisms.
- Minimize: confirmed failures are shrunk to a minimal reproducible counterexample.
- Export: evidence package with manifest, hashes and claim ledger.

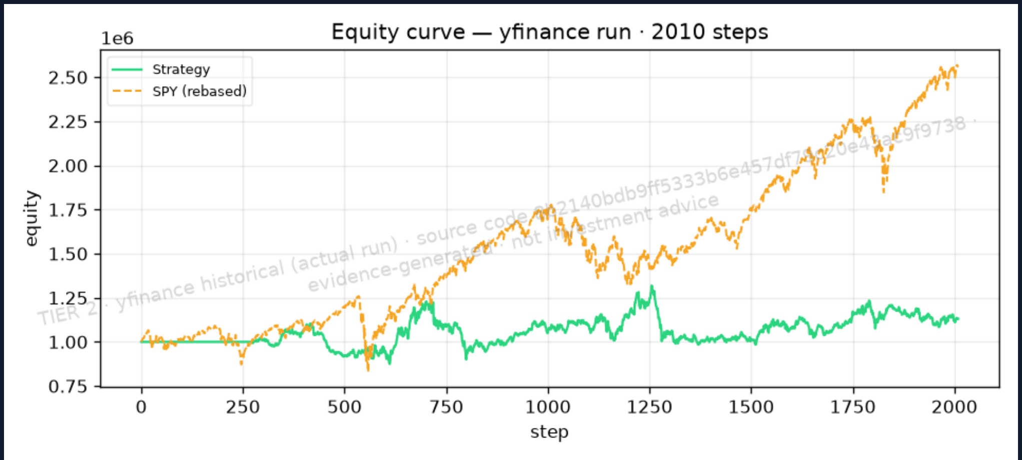


Demo run — Real historical backtest (yfinance)

30 assets · 50 trades · all figures from deck_data.json

13.07%	0.18	-26.75%	14.65%	14.03%	50
cumulative return	Sharpe	max drawdown	ann. volatility	cost % of capital	trades

- CAGR 1.55% · Sortino 0.22 · Calmar 0.06
- Benchmark CAGR 12.49% · information ratio -0.46
- Gross exposure avg 103.70% · net exposure avg 0.00%
- Costs are explicit bounded heuristics (commission/spread/slippage/borrow) — not broker calibrations.



What the normal backtest missed

Baseline vs confirmed stress failure

- Baseline (Real historical backtest (yfinance)): Sharpe 0.18, max drawdown -26.75%, cumulative return 13.07%.
- Sealed stress search: 8 confirmed failures out of 24 worlds (33.33% failure rate).
- Confirmed-failure mechanisms: borrow_cost_increase, correlation_breakdown, slippage_inflation, volatility_expansion.
- Stress-mechanism result matrix (per mechanism, base seed):
- momentum_reversal: pass
- volatility_expansion: FAIL (low_sharpe)
- correlation_breakdown: FAIL (low_sharpe)
- volatility_compression: pass
- spread_inflation: pass
- slippage_inflation: FAIL (low_sharpe)
- borrow_cost_increase: FAIL (low_sharpe)
- short_unavailability: pass
- delayed_rebalance: pass
- universe_churn: pass
- missing_data_shock: pass

Minimized counterexample

Smallest reproducible world where the locked strategy still fails

• Baseline vs minimized failure — the breaking condition, shrunk to a minimal reproducible counterexample:

• • BASELINE (Real historical backtest (yfinance)): Sharpe 0.18, cum. return 13.07%.

• • MINIMIZED FAILING WORLD (volatility_expansion @ intensity 0.9285, seed 20264): still fails = True; violated predicates = low_sharpe.

• • Adjacent pass: no adjacent pass within radius — reported honestly, not fabricated.

Education & research value

A falsification lab, not a profit promise

- Students see WHY a strategy fails — mechanism, intensity, seed — not just an equity curve.
- Every claim is bound to a hash and an evidence file; grading and review become reproducible.
- Instructors can replay the minimized counterexample deterministically.
- Mechanism library searched this run: borrow_cost_increase, correlation_breakdown, delayed_rebalance, missing_data_shock, momentum_reversal, short_unavailability, slippage_inflation, spread_inflation, universe_churn, volatility_compression, volatility_expansion.

Architecture & moat

Immutable contracts + sealed synthetic worlds

- Clause ledger → canonical hash: the same immutable artifact flows through every stage.
- Sealed synthetic world generator: seeds + mechanisms are reproducible and tamper-evident.
- Evidence packages carry SHA-256 manifests binding data, backtest, campaign and replay.
- Failure minimizer + adjacent-pass search turn red flags into actionable counterexamples.
- Moat: the corpus of confirmed, minimized failures compounds with every run.

Honest boundary & roadmap

What this is NOT — and what comes next

- Synthetic stress worlds are generated, not historical; they probe fragility, not real future risk.
- Costs are explicit bounded heuristics (flat bps), not validated broker calibrations.
- Fenrix fundamentals (if used) are not point-in-time; lagged approximation only.
- No claim of live profitability, production execution fidelity, or universal realism.
- Failure search is not exhaustive; it reports confirmed failures only.
- Roadmap: point-in-time fundamentals, broker-calibrated costs, richer mechanism library, live paper-trading bridge.

Ask

Help us make strategy falsification the default

- Pilot with quant-finance classrooms and student funds this semester.
- Feedback on mechanism realism from practitioners.
- Compute credits for larger sealed stress campaigns.
- Reproduce everything: make submission-demo · make verify-submission · make pitch-deck.